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Rotating Plasma Experiments. II. Energy Measurements and the Velocity Limiting Effect

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Rotating Plasma Experiments. II. Energy Measurements and the Velocity Limiting Effect

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The total rotational energy and energy loss rate in Ixion III, the Los Alamos rotating plasma experiment, have been measured as functions of time. As the radial electric field is increased, a limiting velocity of rotation of approximately 1.3×10^7 cm/sec is observed. Measurements indicate the presence of a region of fully ionized plasma whose contaminants radiate large amounts of energy in the form of line radiation in the vacuum ultraviolet. Mechanisms that have been proposed to explain the velocity limit are discussed in relation to the experimental results.

I. INTRODUCTION

THE preceding paper[1] on the Ixion experiment at Los Alamos describes the general physical operation of this rotating plasma device. In the present paper results are given of experiments which permit the rotating plasma to be disconnected from the charging capacitor bank and its energy content and losses to be measured. This technique is further exploited when various loads and high-voltage sources are applied subsequent to the disconnection.

The application of high voltages to the rotating plasma shows the existence of a limit on the rotating velocity which can be maintained without violent voltage breakdown. Current-measuring loops introduced into the interior of the chamber confirm that this limit is indeed a characteristic of the plasma and is not controlled by voltage breakdown at the boundary insulators.

An indication of the nature of the process which limits the rotational velocity is obtained by means of observations on contaminant line radiation in the vacuum ultraviolet.

II. FREEWHEELING OPERATION

In the usual operation of Ixion the rotation of the plasma is maintained by the energizing capacitor bank for a time which is governed by the capacitance of the bank and the internal resistance of Ixion. A technique has been developed whereby the energizing bank can be disconnected from Ixion after rotation is established. In this way the natural decay of the rotational motion of the plasma can be observed. The operation after removal of the energizing bank is designated by the term "freewheeling."

[1] D. A. Baker, J. E. Hammel, and F. L. Ribe, Phys. Fluids 4, 1534 (1961).

The experimental method is indicated in the circuit diagram of Fig. 1. A molybdenum-anode ignitron S_1 switches the energizing capacitor bank C_0 onto Ixion. After the charging current has passed, ignitron S_2 is triggered and shorts the energizing bank.

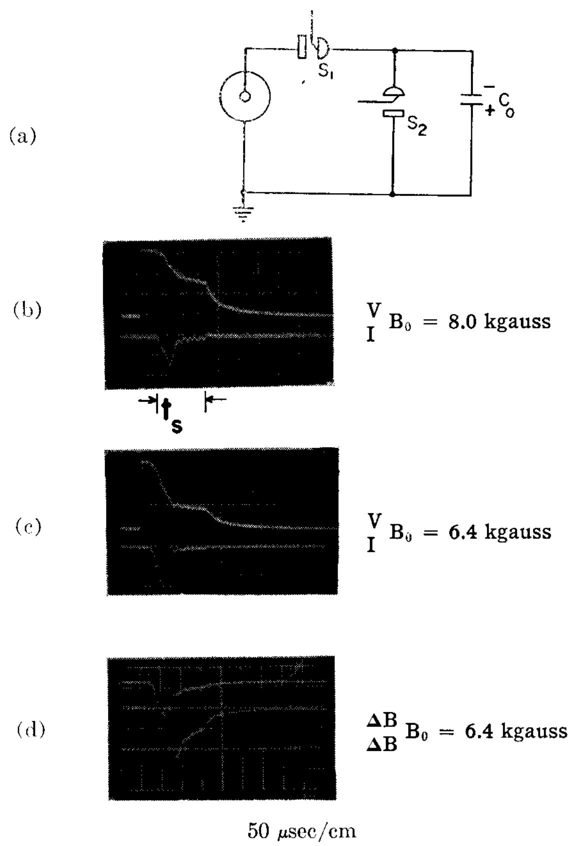


FIG. 1. (a) Circuit diagram for freewheeling operation of Ixion. (b) and (c) Voltage-current oscillograms for freewheeling operation. Voltage scale, 2 kv/div; current scale, 15 kamp/div; (d) Diamagnetic probe signal ΔB on axis at the midplane. Upper trace, 0.8 kgauss/div; lower trace, 0.2 kgauss/div.

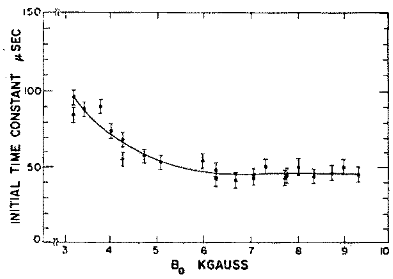


FIG. 2. Ixion voltage decay time constant. Capacitor bank disconnected 160 μsec after application of radial electric field. Initial capacitor voltage = 7.0 kv.

The voltage across S_1 is then reversed, causing rectification in this ignitron and isolation of Ixion. The subsequent decay of the voltage on Ixion is a measure of the decay of the motion of the freely rotating plasma. The experiment depends upon the rectifying properties of water-cooled metal-anode ignitrons.

Figures 1(b) and 1(c) show Ixion's voltage and current during the complete sequence of operations for two values of applied magnetic field B_0, measured at the midplane. During the freewheeling the voltage decays exponentially with time.

The time constant of the voltage decay is measured as a function of B_0 and is shown in Fig. 2. Its value remains constant except at extremely low magnetic fields. The time constant is also determined as a function of t_s, the time between the onset of the charging current pulse which energizes the plasma and the start of freewheeling. The decay time is found to be very nearly constant (between 35 and 47 μsec) for all values of t_s between 50 μsec and 1 msec.

Figure 1(d) is an oscillogram of a time-integrated axial magnetic probe signal taken at the center of Ixion. Analysis of the probe data during freewheeling shows that the signals decay at the same rate at all radial positions. The probe signal at the center throughout the decay time is proportional to V^2/B_0^3. This is the same relationship found for the "holding mode" (sustained back emf) as described in paper I. This property of the diamagnetic signal provides evidence that the number of rotating ions does not change appreciably during the freewheeling decay period.

III. MEASUREMENT OF THE ROTATIONAL ENERGY

The experiment now to be described is concerned with measuring the time dependence of the ordered rotational energy stored in Ixion while operating in the voltage-holding mode. In principle, the stored energy at any prescribed time can be obtained from a measurement of the energy delivered to a low-resistance load connected by means of a switch between the machine electrodes. Subtraction of the bank energy $\frac{1}{2}C_0V^2$ (evaluated at the time of load connection) from the total energy dissipated in the load, gives the recoverable energy stored in the rotating plasma. The accuracy of this technique is greatly enhanced by disconnecting the main bank (by the method described above) which contains over 90% of the energy, just prior to connecting the resistance load. In this way the total energy delivered to the load must come entirely from ordered motion in Ixion. In the arrangement shown in Fig. 3, ignitron S_3 is activated 2 μsec after S_2 has initiated the freewheeling.

Low values for load resistance are desirable so that a large portion of the energy goes into the load rather than into the internal losses. As a result of the finite length of time required to draw energy from the plasma, the total energy in rotation is always somewhat greater (of the order of 20%) than that delivered to the load in the experiment. In practice, the minimum load resistance is about 2 ohms, since lower values cause the circuit in-

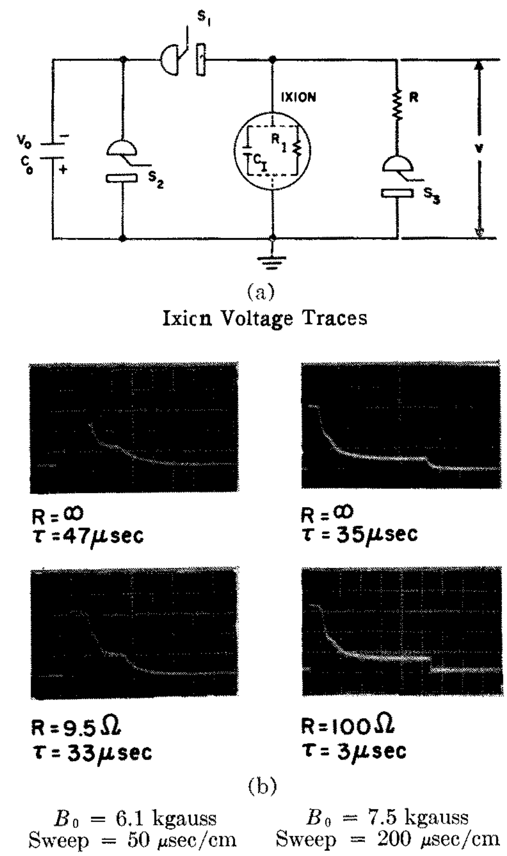


FIG. 3. Circuit for loading rotating plasma and voltage traces for various loads and load application times. Voltage = 2.2 kv/div. τ is the time constant for decay of the voltage.

ductance to become important and complicate the energy measurements. In addition, if the voltage on Ixion is decreased too rapidly (before the shorted bank voltage has decayed) the voltage will reverse on S_1, causing it to conduct once again.

In order to take account of the internal dissipation losses, the Ixion device is represented by the equivalent circuit shown in the circle of Fig. 3. The Ixion capacitance and resistance $C_I(t)$ and $R_I(t)$ characterize the device, and the instantaneous stored rotational energy and internal power dissipation are given by $\frac{1}{2}C_I V^2$ and V^2/R_I, respectively. The parameters R_I and C_I were deduced by measuring two decay times τ_1 and τ_2 for two different load resistors $R = R_1$ and $R = R_2$ (cf. Fig. 3). This technique made use of the fact that the Ixion voltage decay under load conditions is always accurately an exponential function of time as shown in Fig. 4. Sample oscilloscope traces under different load and switching times are shown in Fig. 3.

The data were analyzed in the following manner: During the time of voltage decay when S_1 is nonconducting, an equation for the circuit in Fig. 3 is

$$[d(C_I V)/dt] + V/R_p = 0 \qquad (t \geq t_s), \tag{1}$$

where R_p is the value of the Ixion and load resistances in parallel and t_s is the time of load connection. Substituting into Eq. (1), the experimentally determined voltage relation

$$V = V_0 e^{-t'/\tau} \qquad (t' \geq 0), \tag{2}$$

yields

$$\frac{dC_I}{dt} - \frac{C_I}{\tau} + \frac{1}{R_p} = 0, \tag{3}$$

where t' is the time measured from the instant the load is connected. Two such equations may be written, each for a different value of load resistance R_1 and R_2. Particle confinement is probably not immediately affected by the load-switching operation since the voltage is continuous, and it will be

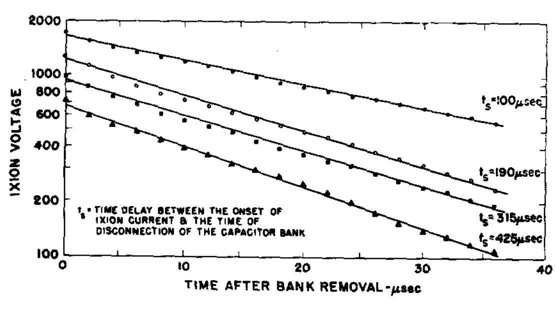


FIG. 4. Ixion voltage decay with 9.5-ohm external load resistance. $V_0 = 7.0$ kv, $B_0 = 6.1$ kgauss.

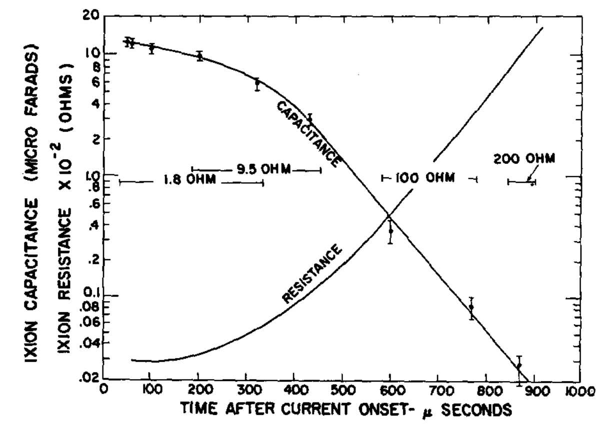


FIG. 5. Capacitance and resistance of Ixion as functions of time after current onset. $B_0 = 7.5$ kgauss. Initial capacitor voltage 7.0 kv.

assumed, therefore, that all capacitance time derivatives are continuous at $t = t_s$. With this assumption, a simultaneous solution of the pair of equations yields the following expressions for C_I and R_I at $t = t_s +$.

$$C_I = \frac{\tau_1 \tau_2 (R_1 - R_2)}{R_1 R_2 (\tau_1 - \tau_2)}, \tag{4}$$

$$R_I = \frac{R_1 R_2 (\tau_1 - \tau_2)}{R_1 \tau_2 - R_2 \tau_1 - R_1 R_2 (\tau_1 - \tau_2)\, dC_I/dt}. \tag{5}$$

Having computed C_I from Eq. (4) for various switching times t_s, the slope dC_I/dt of the resulting curve is substituted into Eq. (5), and the value of R_I is calculated. With the additional assumption that R_I remains unchanged during the switching operation, these values found for C_I and R_I are also valid for the conditions immediately before the switching time. The results of the computations are given in Fig. 5. In the calculations the value used for R_1 is indicated by the brackets in the figure and R_2 was infinite. The following facts support the assumptions made in the calculations: (1) At points where data were taken with two different values of R_1, the computed values of Ixion resistance and capacitance agree. (2) A calculation using values of C_I and R_I given in Fig. 5 reproduces the experimental voltage curve of Ixion.

It is interesting to note that although R_I and C_I vary over three powers of 10, the product of R_I and C_I (decay time) remains nearly constant over a time interval of a millisecond.

The rotational energy $\frac{1}{2}C_I V^2$ can be computed at any time from the instantaneous values of Ixion capacitance and voltage. The results of this calculation, showing energy as a function of time for $B_0 = 7.5$ kgauss, are shown in Fig. 6.

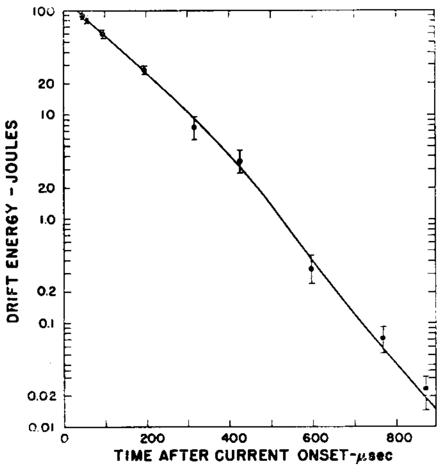
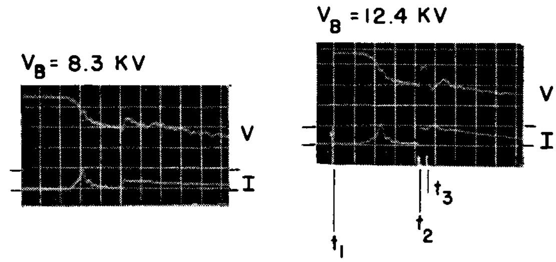


FIG. 6. Ixion rotational energy as function of time after current onset for B_0 = 7.5 kgauss. Initial capacitor voltage 7.0 kv.



(a) (b)

Sweep = 50 μsec/cm

FIG. 8. Voltage and current traces for two values of boost voltage V_B. Tick marks indicate the positions of the voltage and current base lines. B_0 = 8.4 kgauss. Voltage scale, 2 kv/div; current scale, 15 kamp/div.

IV. VOLTAGE LIMITING IN IXION

Ixion is normally set into rotation by the energizing capacitor bank, charged to 7.5 kv. If the bank voltage is raised appreciably above 7.5 kv, the internal resistance becomes very low and the sustained back emf is nearly zero. A two-step experiment was performed to increase the voltage after ionization and rotation were well established. The plasma was set into rotation in the usual way with the main capacitor bank C_0 charged to 7.5 kv. Then after the rotation was established, the main bank was disconnected and a second bank C_B, called the boost bank, was immediately switched onto Ixion in its place. The circuit diagram for the operation is shown in Fig. 7. The boost-bank voltage V_B could be made as large as 20 kv, and provision was made for inserting a series impedance to allow control of the rate of increase of Ixion's voltage. All the data to be presented in this paper were taken with a 1.33-ohm series resistor in the boost-bank circuit. Oscilloscope traces of the Ixion current and voltage under boost conditions are shown in Fig. 8. Figure 8(a) gives oscilloscope traces for an initial boost-bank voltage of 8.3 kv, while Fig. 8(b) corresponds to 12.4 kv. The time of connecting the main bank is designated by t_1 and the time of connection of the second bank by t_2. At the lower boost voltage [Fig. 8(a)] one can see that Ixion's volttage is increased above its steady value by about 1 kv and then decays as the boost capacitor is discharged. In the case of the higher boost voltage, the voltage rises about 2 kv and levels off. At a later time t_3, however, the impedance of the machine suddenly decreases, causing an increase of current and a rapid decrease of the Ixion voltage. In both cases the Ixion impedance adjusts itself in such a way that the peak voltage attained remains far below the boost-bank voltage with the difference in the voltages appearing as a potential drop across the series resistor. It was found that even with boost-bank voltages as great as 20 kv, the peak Ixion voltage never exceeded 8 kv.

In order to determine what role the end insulators play in the effects just discussed, an experiment was performed to measure the radial currents flowing across the end insulator surfaces (cf. Fig. 2 of paper I). An internal current-measuring loop was installed on each insulator, one of which is shown in Fig. 9. The loop was a glass enclosed Rogowski coil of 10-cm major diameter, located 2 cm from the insulator surface. Since the loop measures the total current threading it, one can compare its

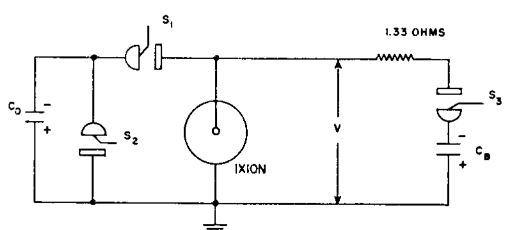


FIG. 7. Circuit for voltage boost experiment.

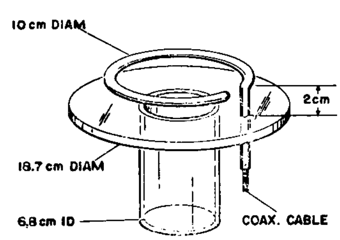


FIG. 9. Glass enclosed Rogowski loop mounted on an end insulator.

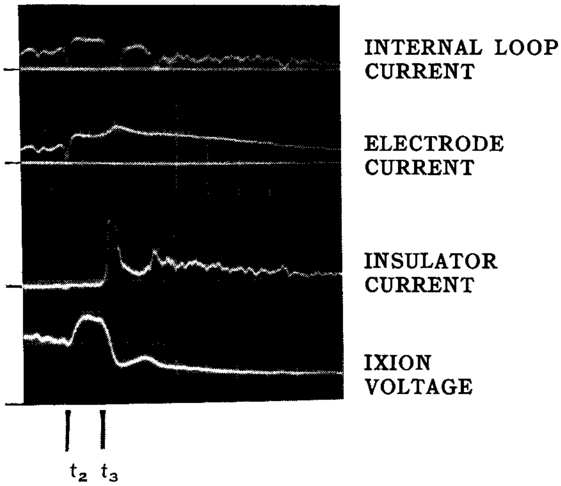


FIG. 10. Oscillograms of loop current, insulator current, electrode current, and Ixion voltage. Boost bank voltage = 12.4 kv, B_0 = 8.4 kgauss. Voltage scale, 2 kv/div; current scale 10 kamp/div.

measurement with the total electrode current measured externally and determine what radial current, if any, flows in the gap between the loop and the insulator. Sample results of the experiment are shown in Fig. 10. The top trace is the time integrated signal of an internal loop, giving the current threading it. The second trace shows the total current supplied to the corresponding electrode. The third trace is the difference of the first two currents, measured by means of a differential amplifier, and shows the time history of the current flowing across the insulator. The bottom trace is the Ixion voltage. From traces such as these, as well as magnetic probe data, one concludes the following: (a) There are no insulator currents during operation with low boost voltages. (b) The peak voltage attained is not controlled by insulator conduction. (c) The abrupt drop of voltage and increase of electrode current at the later time t_3 is a result of the onset of insulator currents.

The sharp impulse of insulator current at the time t_3 is attributed to the discharge of a part of the previously established rotating energy when the voltage drops as a result of the sudden breakdown in the insulator region. The reason that the onset of insulator current is delayed for some time beyond the voltage peak is presently unknown. The fact that the delay occurs over the complete range of boost voltages investigated leads to the conclusion that the peak voltage obtained is controlled by the plasma rather than by conduction effects at the insulator surfaces.

The dependence of peak voltage on magnetic field was investigated and the results appear in

FIG. 11. Peak Ixion voltage attained under boost with magnetic field as a parameter.

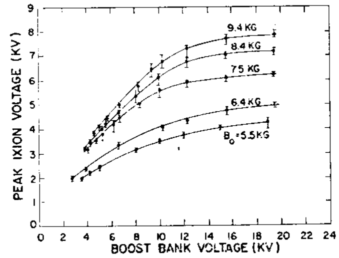


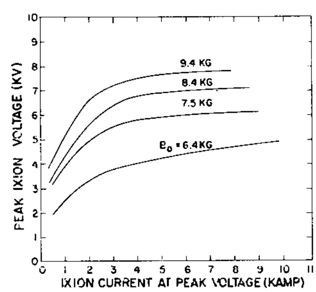
Fig. 11. The abscissa is the initial voltage of the boost bank, and the ordinate is the corresponding peak Ixion voltage which was attained. Each curve was obtained with a different value of applied magnetic field B_0. It is apparent that the machine displays a strong voltage saturation and that the value of the ratio of saturation voltage to magnetic field is very nearly constant at 0.83 v/gauss. This corresponds to a limiting velocity of 1.3×10^7 cm/sec when the electric field is estimated by scaling the electrostatic-probe data given in paper I. The current-voltage characteristics of Ixion for the same set of operating conditions are shown in Fig. 12. The limiting Ixion voltage is plotted as a function of the corresponding Ixion current, with the magnetic field as a parameter.

The diamagnetic signal on the axis during the voltage boost operation was measured in the way described in paper I. The V^2/B_0^3 dependence of the magnetic signal ΔB is the same as the unboosted and freewheeling result. This provides strong evidence that the number of ions remains constant for at least the duration of the measurement which included the first 75 μsec of the boosted operation.

V. ULTRAVIOLET RADIATION AND THE ENERGY LOSS FROM IXION

The first experiment to be described in this section was performed in an attempt to determine the approximate degree of ionization of the gas in Ixion. When it became apparent that the gas was nearly fully ionized, the last three were performed to determine if the contaminant radiation represented an important plasma energy loss.

FIG. 12. Ixion current-voltage characteristics with magnetic field as a parameter.



The four experiments to be discussed are the following:

(1) The use of a vacuum ultraviolet monochromator to search for line radiation from contaminants such as C, N, and O.

(2) Measurement of the total energy density falling on the wall as a function of time by observing the temperature rise of a thin molybdenum foil.

(3) Measurement of the total energy carried to the wall by photons by means of a pinhole camera using Eastman SWR film.

(4) A photocell measurement of the incident photon flux which yields the mean photon energy when combined with the results of (2) and (3).

A. Results from uv Spectrum Analysis

A Jarrell-Ash Seya-Namioka vacuum spectrometer was placed to detect radiation through a central viewing port at the midplane of Ixion, and a survey was taken using the instrument as a spectrograph with Eastman SWR film. The survey showed prominent lines throughout the range of the instrument from 400 to 4000 A. The small dispersion of the instrument and the poor focus at positions other than the center of the film made identification very difficult for most of the lines except the prominent 1548 A doublet of C IV and the corresponding doublets in N V and O VI. In addition to the spectrographic survey, the instrument was used as a

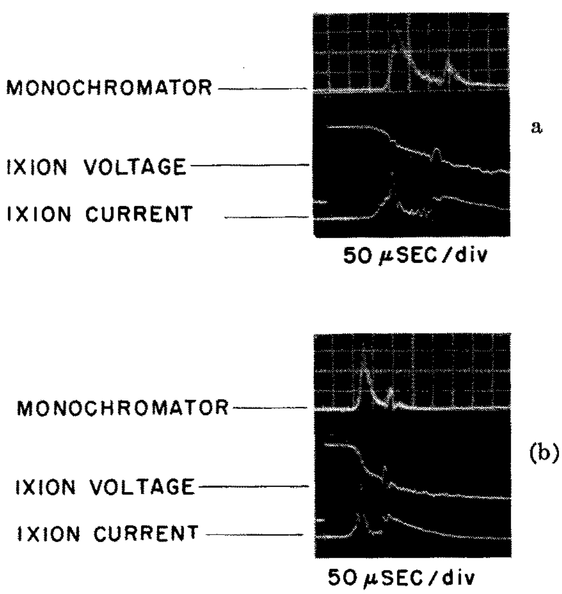


FIG. 13. Monochromator output when set at 1548 A with the corresponding Ixion current and voltage traces. (a) Normal operation; (b) Current breakdown along insulator after boost. The sudden drop in light intensity corresponds to the discharge leaving the midplane region.

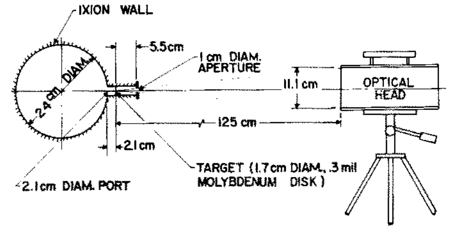


FIG. 14. Experimental arrangement for the bolometer measurements.

monochromator with a photomultiplier detector. The time-resolved light output of the plasma was observed over a large region of the uv spectrum. Positive identification was made of the C IV doublet at 1548 A, the corresponding N V doublet at 1238 A, and the O VI doublet at 1032 A. The O VI line became less intense as the interior surfaces of Ixion were cleaned with successive discharges. The N V line was observed only when N_2 was introduced as a contaminant in the injected gas. A search for prominent lines from lower ionized states of these contaminants revealed nothing above the background signal.[2] The above evidence indicates that a major portion of the plasma is fully ionized. Since the ionization potential of O V is 138 ev, the presence of O VI implies an electron temperature of approximately 18 ev or greater, as deduced from the coronal equation.[3,4]

Many of the large number of lines observed throughout the spectrum seemed to correspond to reported Ni and Fe lines; however, no positive identification was made. The presence of these elements might be suspected because of the stainless-steel electrodes.

Oscillograms of the output of the monochromator when set on the 1548 A C IV line, as well as current and voltage, are shown in Fig. 13(a). A boost-bank voltage of 10 kv was applied to Ixion 150 μsec after the current onset. The monochromator output is approximately proportional to the power input to Ixion. Careful analysis shows, however, that the peaks of the light signal occur from 6 to 10 μsec after the peaks of the power. An interesting feature

[2] The presence of C II radiation was reported previously [*Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy* (United Nations, Geneva, 1958), Vol. 31, p. 319]. This was observed from visible spectra taken with a monochromator looking into an off-axis port. This only permitted a tangential view of the outer 1 cm of the chamber. The radiation from the cooler outer layer was probably masked when the monochromator was viewing the intense central portion of the discharge.

[3] V. G. Elwert, Z. Naturforsch. **7a**, 432 (1952).

[4] R. F. Post, in *Annual Reviews of Nuclear Science*, edited by E. Segrè and L. I. Schiff (Annual Reviews, Inc., Palo Alto, California, 1960), Vol. 9.

FIG. 15. Oscillograms of infrared detector output and Ixion voltage. Voltage boost 100 μsec after current onset. Sweep speed = 50 μsec/div.

of the monochromator signal is an abrupt drop in intensity when there is a sudden decrease in the voltage after boost. This is shown in Fig. 13(b) and is taken to be a result of the discharge current transferring to the insulator region, as found in the internal current-loop experiments described above.

B. Bolometer Measurement of Energy Flux

The energy flux to the wall at Ixion's midplane was measured absolutely with time resolution by observing the temperature rise of a thin Mo disk. The geometry of the experiment is shown in Fig. 14. A carbon-blackened Mo foil target having a thickness of 7.6 μ was positioned near the outer wall in a glass tube so as to intercept a portion of the escaping energy. Its temperature rise during the discharge was then measured as a function of time by means of an infrared detector.

Typical oscilloscope traces of the detector output under boost conditions are shown in Fig. 15. Taking into account the area of the Mo disk and making an approximate solid-angle correction, the energy densities detected at the Ixion wall after unboosted and boosted discharges are found to be 1.6 and 2.2 joules/cm^2, respectively.

The temperature rise of the Mo disk and the power and energy input from the capacitor bank to Ixion are shown as functions of time in Fig. 16 and 17. Note that, except for a constant scale factor the temperature rise and energy input curves are nearly the same. Below we shall compare the absolute energy input with that observed at the wall from a consideration of the plasma geometry.

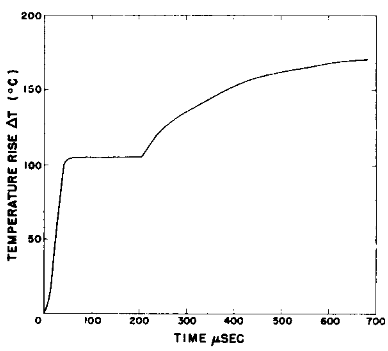


FIG. 16. Temperature of Mo disk as deduced from infrared detector signal.

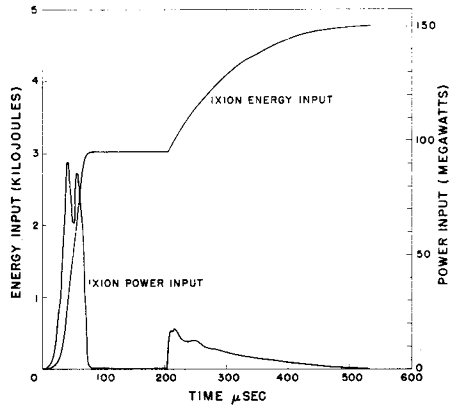


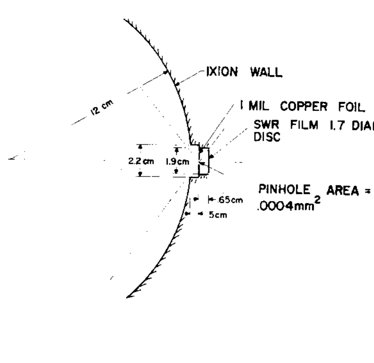
FIG. 17. Power and energy input to Ixion corresponding to the temperature curve shown in Fig. 16.

Since lithium fluoride transmits radiation of wavelengths greater than 1200 A, a LiF window was placed over the disk in order to determine the portion of radiated energy carried by long wavelength photons. A small signal of only a few percent of that observed without the window was observed. This means that radiation above 1200 A cannot account for more than a few percent of the energy flux to the Ixion wall. After a few discharges the signal through the window disappeared. This was probably due to the formation of a coating on the LiF window. A visible coating was indeed present when the window was removed.

C. Energy-Flux Measurements by Means of Pinhole Photographs

In order to determine the energy carried to the wall by photons, a pinhole photograph of the discharge was taken using Eastman SWR low-gelatin film in the configuration shown in Fig. 18. A control film placed outside the light path indicated that no blackening was being caused by atomic hydrogen. An uncertainty exists in this measurement owing to the lack of precise sensitivity data at wavelengths between the soft x-ray region and 1216 A. However, there are indications that the sensitivity in this

FIG. 18. Pinhole camera arrangement.



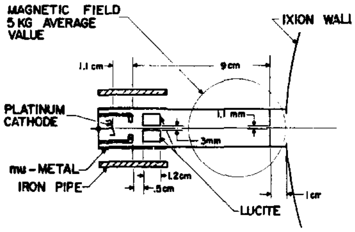


FIG. 19. Schematic diagram of apparatus for photocell measurement.

range of wavelengths is constant to within $\pm 50\%$ of that at 1216 A.[5,6] Measurement of the exposure to one discharge (unboosted) gave a density of 10 ergs/cm^2 at the film. This corresponds to about 0.5 joule/cm^2 at the wall and compares reasonably well with the bolometer value of 1.2 joules/cm^2. The presence of the 8.5 kgauss field prohibits deuterons having energies less than 100 ev from reaching the film.

D. Photocell Measurements

The fourth detector used to measure the radiation from Ixion was a photocell having an untreated Pd cathode. This detector is sensitive to uv radiation having wavelengths less than 1000 A. Figure 19 shows a cross-section diagram of the detector system at the midplane port of Ixion. The presence of the magnetic field and the small angular acceptance served to eliminate charged particle effects. A bias voltage of -35 v was placed on the photo cathode during the data taking. This biased the photocell well past its saturation voltage of -15 v.

The power input of the discharge (VI) is plotted with the photocell current in Fig. 20. The photocell current follows the power input but with a buildup time delay of 5 to 10 μsec. This delay may be associated with the time required to populate the higher ionized states of the contaminants.

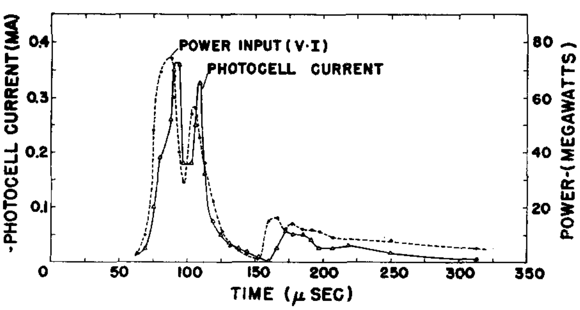


FIG. 20. Photocell current and power input to the discharge as functions of time.

When the total integrated photocell current is compared with the energy measured by the Mo disk and the uv camera, an average photon energy of at least 50 ev is deduced when a photoelectric conversion efficiency of 0.08 is assumed. Such photon energies would be expected from the highly ionized contaminants whose longer wavelength lines were observed in the discharge by means of the vacuum monochromator. The 0.08 photoelectric conversion efficiency used in obtaining the average photon energy is a reasonable interpolation between the values given by Weissler[7] up to 26 ev and the values in the x-ray region.[8]

It should be mentioned that the integrated photocell current could also be accounted for by a large flux of photons in the 10 ev region where the conversion efficiency for the photocell is low. This possibility is eliminated by the monochromator results and the LiF transmission experiment.

E. Comparison with an Isotropic Source

The results from these detectors may be compared with the energy density which would be incident on the wall if the emission were from an isotropic plasma source. For this comparison the radiation is assumed to come from a homogeneous volume with the dimensions of the plasma as found by the magnetic probes (cf. paper I). The known input of $\sim$3000 joules (unboosted operation) then corresponds to an energy density of 1.5 joules/cm^2 at the midplane wall. This energy density falls well within the range of uncertainty of the measured value. This result and the fact that the time resolved signals from the monochromator, the bolometer, and the photocell are essentially coincident in shape with the power input implies that a large portion of the energy is lost from the discharge in the form of hard uv radiation.

VI. DISCUSSION

The characteristics of the ordered motion of the plasma have been established by the magnetic probe measurements of paper I and the freewheeling and loading experiments. The drift velocity (8×10^6 cm/sec), the density (1×10^{-8} g/cm^3), and the active volume (2 liters) as found with the electrostatic- and magnetic-probe measurements (paper I) give a value for the ion drift energy which is within 15% of the 80 joules in rotational energy measured

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at the corresponding time ($\sim$65 μsec after current onset) in the freewheeling and loading experiments. The capacitance measurements indicate an ion density which is nearly constant for 100 μsec, again supporting the result from the magnetic probe measurements of the diamagnetism.

The internal rotational energy loss is characterized by the resistance $R_{\rm I}$ in the equivalent circuit for Ixion (cf. Fig. 3). In the boosting experiment it is seen that the loss rate can become arbitrarily large in order to maintain the characteristic limiting value of drift velocity. Until the radiation measurements had been made on Ixion, this energy loss was taken as an indication of the presence of neutral atoms in the plasma as a cause of the loss.[9,10] According to the ion-neutral interaction hypothesis, the energy is carried out by neutral particles which gain energy by collisions with rotating ions. Such a model gives a radial current which is proportional to the ion density and thus a resistance $R_{\rm I}$ proportional to $1/n$ (n = ion density). This dependence is required under the usual operating conditions since it is found experimentally that $R_{\rm I} \propto 1/C_{\rm I}$ and the capacitance is of course proportional to n. The increased loss rate when the voltage limit is approached would be explained under the ion-neutral hypothesis by a rapid rise in the ion-neutral collision cross section.

A limit to the velocity of ions passing through a neutral gas was predicted by H. Alfvén.[11] This velocity limit was predicted to occur when

$$\tfrac{1}{2}Mv_c^2 = eV_{\rm ion}. \qquad (6)$$

Here v_c is the limiting velocity, $eV_{\rm ion}$ is the ionization energy, and M is the ion mass. The rotating plasma experiment of Block and Fahleson[12] shows the existence of such a limiting velocity. In that experiment the power input to the discharge is always limited by an external resistance to the extent that partial ionization is ensured. A difficulty is encountered in this ion-neutral theory, since it is only for electrons that the ionization cross section becomes large when the kinetic energy slightly exceeds the ionization energy of the neutral. Allis[13] considered this problem in connection with the energy loss in Ixion. He has shown that electrons rotating through a background of neutral gas will gain random energy at each elastic collision with a neutral until their random energy equals the energy of the neutral gas as seen in the frame of the moving electron. A limiting velocity of rotation would be reached when a sufficient number of electrons have attained high enough energies to produce an appreciable neutral ionization rate. In the process of gaining energy the electrons will remain essentially Maxwellian; therefore it is not clear why the limiting velocity should be given by Eq. (6). An electron temperature of $\sim(1/10)\ eV_{\rm ion}$ would be sufficient. In Ixion, however, the velocity limit corresponds to a deuteron energy of more than 170 ev or about 11 times the energy of the velocity limit found by Block and Fahleson. The experimental evidence indicates that the electron temperature is in excess of 18 ev and that therefore a major portion of the plasma is fully ionized. Thus ion-neutral interactions in the body of the plasma cannot account for all of the energy loss and the velocity limit in Ixion.

Simple viscosity effects based on Coulomb collisions between ions, as proposed by Baker, Bratenahl, De Silva, and Kunkel,[14] for the loss mechanism in the homopolar experiment are of no consequence in the operation of Ixion. Calculations of the viscous transport of momentum across the magnetic field show that the outer electrode could not have a noticeable drag effect in any such simple way. Since no currents were found in the vicinity of the end insulator, viscous effects along the field direction can not be responsible for the velocity limit.

In view of the large amount of energy emerging from the plasma as uv radiation, it is likely that this energy loss is the dominant one. The voltage limiting would be associated with the excitation by electrons of levels in highly ionized contaminants such as C IV, O VI, etc. A large power loss in the form of far uv radiation is not unique to the Ixion experiment. Such a loss has also been observed in the plasma of a toroidal discharge with similar electron temperature and density by Karr *et al.*[15] (Perhapsatron S-4).

In postulating uv radiation as the means of velocity limiting it is important to note that the photons are capable of transporting the energy,

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but not the momentum, directly from the plasma to the outer wall of Ixion. At present the means of momentum transport is not known. A reasonable mechanism would be that of greatly enhanced viscosity arising from plasma turbulence brought about by instabilities associated with the high-velocity flow. Such instabilities could also account for the transformation of ordered ion rotational energy to the random energy of the electrons necessary to excite the impurities.

From the electrostatic probe measurements (paper I) it appears that energy is deposited in the plasma well away from the electrodes and walls. The means by which this energy is converted into high electron temperature and the exact nature of the internal resistance are problems of great interest and will be subjects of future investigations. It should be emphasized that the internal resistance R_I governs to a large extent the character of the rotating plasma by determining the radial electric field and thus overriding any plasma polarization sheath effects that might otherwise be expected to control the radial distribution of electric potential.

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High-β Injection into a Magnetic Mirror Field

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Axial injection of a high-density helium plasma into a magnetic mirror has been experimentally studied. Observations of the plasma-field interaction were made with magnetic probes, electrostatic probes, piezoelectric probes, and an optical monochromator which analyzed emission-line profiles. In the central plane of the mirror a density of $2 \pm 1 \times 10^{15}$ ions/cm^3 and a *maximum* ion temperature of 10 ev are indicated. In the upstream region ion temperatures of 35 ± 5 ev were recorded. Diamagnetic signals show that the high-β plasma "stagnates" near the upstream throat of the mirror. Also, the derived plasma pressure measured on axis decreases exponentially downstream. Piezoelectric-probe measurements show large axial-momentum transfer rates in the downstream region of the mirror, where no diamagnetic signals are observed. The data are compared with a fluid model of a plasma which describes the injection, "stagnation," and effusion along field lines. This experiment indicates that the axial injection of a collision-dominate, high-β plasma into a static mirror field is not efficient for producing and confining a high-temperature plasma.

INTRODUCTION

MUCH experimental interest has recently developed in the injection of high-density highly directed plasmas into various vacuum magnetic fields.[1-6] In part, this interest stems from the in-

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